Computability theoretic forcing*

How to prove theorems about Turing reducibility the hard way.

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1 Reducibility, hierarchies, and complete sets

The concept of relative computability goes back to Turing's 1939 paper and is usually modeled using oracle Turing machines. Informally a function $f : \omega \to \omega$ is **computable relative to** $A \subseteq \omega$ if there is an algorithm for computing $f(n)$ where the algorithm has access to elementary information about A, i.e., it may ask questions of the form "is $k \in A$"? during execution (and will also receive answers to these questions).

More formally, an oracle Turing machine Γ is a Turing machine with an additional read-only "oracle tape". The machine has an independent oracle reading head which it may move along the oracle tape and read the bit under it.For an infinite binary string $x \in 2^\omega$, we write Γ^x for the oracle Turing machine with x written on its oracle tape.

A function $f : \omega \to \omega$ is computable relative to $A \subseteq \omega$ if there is (a program for) an oracle Turing machine Γ such that for every n, $f(n) = \Gamma^A(n)$ (strictly speaking we put the characteristic function χ_A of A on the oracle tape and not A. To ease notation we identify A with its characteristic function when appropriate). A set $A \subseteq \omega$ is computable relative to $B \subseteq \omega$ if its characteristic function χ_A is computable relative to B.

Any computation $\Gamma^A(n)$ must terminate after a finite amount of steps and thus can only read a finite amount of the oracle tape. So there must be some $k \in \omega$ so that if $B \upharpoonright k = A \upharpoonright k$, then $\Gamma^B(n) = \Gamma^A(n)$. The least such k is called the **use** $u(\Gamma, A, n)$ of $\Gamma^A(n)$. Motivated by this we also allow finite strings $\sigma \in 2^{<\omega}$ on the oracle tape. We use the convention that if $\Gamma^\sigma(n)$ tries to move past σ on the oracle tape then $\Gamma^\sigma(n) \uparrow$ (i.e., is undefined) and that if $\Gamma^\sigma(n) \downarrow$, then it does so within $|\sigma|$ steps, i.e. $\Gamma^\sigma(n) \downarrow$ iff $\Gamma^\sigma_{|\sigma|}(n) \downarrow$.

The fact alluded to above that if $\Gamma^\sigma(n) \downarrow$ and $A \succ \sigma$, then $\Gamma^A(n) = \Gamma^\sigma(n)$ is called the **use principle**. The use principle is an important property of oracle computations. We will now gather some more facts which follow from easy adaptations of the proofs of similar facts for Turing machines.

Proposition 1.1.

1. *There is a* ***universal oracle Turing machine****, i.e., a Turing machine* Φ *such that for each Turing machine* Γ*, there is an* $e \in \omega$ *such that for all* n *and* $X \subseteq \omega$ $\Phi^X(e, n) = \Gamma^X(n)$*. We will write* Φ_e *for* $\Phi^-(e, -)$ *and refer to* $(\Phi_e)_{e\in\omega}$ *as* ***standard enumeration*** *of Turing machines.*

2. *(*The s-m-n theorem*) There is an injective computable function* $s : \omega^2 \to \omega$ *so that for all* x, y, z *and all* $X \subseteq \omega$ $\Phi^X_{s(x,y)}(z) = \Phi^X_x(y, z)$*.*

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3. *(*The padding lemma*) There is an injective computable function $p : \omega^2 \to \omega$ so that for every n and every i $\Phi_n^X = \Phi_{p(n,i)}^X$.*

4. *(*The recursion theorem*) Let $f : \omega \to \omega$ be a computable function. Then there is some e such that for all X, $\Phi_e^X = \Phi_{f(e)}^X$.*

1.1 Turing reducibility and the Turing degrees

Definition 1.1. Let $(\Phi_e)_{e\in\omega}$ be an enumeration of the oracle Turing machines. We say that A is computable relative to B, or, A is **Turing reducible to** B, $A \leq_T B$, if there is an e such that $\Phi_e^B = A$. A and B are **Turing equivalent**, $A \equiv_T B$, if $A \leq_T B$ and $B \leq_T A$.

The **Turing degree** of A is

$$deg_T(A) = \mathbf{a} = \{B : B \equiv_T A\}.$$

The set of Turing degrees is denoted by $\mathcal{D}_T$. It is partially ordered by the relation

$$\mathbf{a} \leq \mathbf{b} \iff \exists A \in \mathbf{a} \exists B \in \mathbf{b}\ A \leq_T B$$

Clearly there is a least Turing degree consisting of the computable sets, the degree of $\emptyset$, denoted by $\mathbf{0}$.

There are many other reducibilities between sets studied in computability theory such as arithmetic, hyperarithmetic, enumeration, (weak) truth table reducibility, etc. Two very important ones that will be useful to us are the following.

Definition 1.2. Let $A, B \subseteq \omega$, then

1. A is **many-one reducible to** B, $A \leq_m B$, if there is a computable function $f : \omega \to \omega$ so that $n \in A$ iff $f(n) \in B$. The induced many-one equivalence is denoted by $\equiv_m$.

2. A is **one-one reducible to** B, $A \leq_1 B$, if there is a computable injective function $f : \omega \to \omega$ so that $n \in A$ iff $f(n) \in B$. The induced one-one equivalence is denoted by $\equiv_1$.

It is not hard to see that $\leq_1$ and $\leq_m$ are finer than $\leq_T$, i.e. $A \leq_m B$ implies $A \leq_T B$. One-one equivalence is often denoted as the *recursive isomorphism* relation as Myhill's isomorphism theorem shows that $A \equiv_1 B$ if and only if there is a computable bijection $f : \omega \to \omega$ such that $n \in A$ iff $f(n) \in B$.

The **computable join** of $A, B \subseteq \omega$ is

$$A \oplus B = \{2n : n \in A\} \cup \{2n + 1 : n \in B\}.$$

Proposition 1.2. *The Turing degrees $(\mathcal{D}_T, \leq)$ form an upper semilattice with*

$$\mathbf{a} \vee \mathbf{b} = deg_T(A \oplus B).$$

Proof. Fix $A, B \subseteq \omega$ of degree $\mathbf{a}$, respectively $\mathbf{b}$. The functions $n \mapsto 2n$ and $n \mapsto 2n + 1$ are computable and thus $A, B \leq_m A \oplus B$, so $deg_T(A \oplus B) \geq \mathbf{a}, \mathbf{b}$. Suppose $\mathbf{c} \geq \mathbf{a}, \mathbf{b}$. Then for any $C \in \mathbf{c}$, $C \geq_T A, B$ and thus also $C \geq_T A \oplus B$. To see this suppose that $\Phi_{e_0}^C = A$ and $\Phi_{e_1}^C = B$ and consider the function $f : 2^\omega \times \omega \to 2$

$$f(X, n) = \begin{cases} \Phi_{e_0}^X(k/2) & if\, n = 2k \\ \Phi_{e_1}^X(k/2) & if\, n = 2k + 1 \end{cases}$$

This function is clearly computable with an index depending only on e_1, e_2 and thus there is an oracle Turing machine Φ such that $\Phi^X = f(X, -)$. To be more precise, consider the program running program e_0 on even inputs and program e_1 on odd inputs and appeal to the Church-Turing thesis. □

1.1.1 The Turing jump

The most important function between sets in computability theory is the **Turing jump**

$$\mathsf{J}: \begin{array}{l} 2^\omega \to 2^\omega \\ A \mapsto A' = \{n : \Phi_n^A(n)\downarrow\} \end{array}.$$

Theorem 1.3 (Properness of the Turing jump)**.** *For all $X \subseteq \omega$, $X <_T X'$.*

Proof. We first show that $X' \geq_T X$. For each n, construct a program $\Phi^X_{f(n)}$ which halts on any input m if and only if $n \in X$. Then $X \leq_1 X'$ via the computable function f, as $n \in X$ if and only if $\Phi^X_{f(n)}(f(n))\downarrow$ if and only if $f(n) \in X'$.

That $X \not\geq_T X'$ is the straight-forward relativization of the fact that the halting problem is not computable. We give it here to illustrate how many facts about computable functions relativize.

Suppose towards a contradiction that $X' \leq_T X$, then the partial function

$$f(n) = \begin{cases} 1 & n \notin X' \\ \uparrow & n \in X' \end{cases}$$

is partial computable from X, i.e., there is an index e such that $\Phi_e^X = f$. But then

$$\Phi_e^X(e)\downarrow \iff e \notin X' \iff \Phi_e^X(e)\uparrow,$$

a clear contradiction. □

Next we see that the Turing jump is degree invariant in a strong sense and thus can be viewed as a function on the Turing degrees.

Theorem 1.4. *For all $A, B \subseteq \omega$, $A \leq_T B \iff A' \leq_1 B'$.*

Proof. Suppose that $A \leq_T B$, via Φ_e and for each n construct a program $\Phi^B_{f(n)}$ such that for any m

$$\Phi^B_{f(n)}(m)\downarrow \iff \Phi_n^{\Phi_e^B}(n)\downarrow.$$

Then $n \in A'$ iff $\Phi^B_{f(n)}(f(n))\downarrow$ iff $f(n) \in B'$, showing that $A' \leq_1 B'$.

On the other hand assume that $A' \leq_1 B'$ via f. Note that relativizing the completeness of $\emptyset'$, if $C \leq_1 B'$, then C is c.e. in B. Hence, A is c.e. in B. To finish we need to show that $\omega \setminus A$ is c.e. in B. To see this consider g such that for all m $\Phi^A_{g(n)}(m)\downarrow$ if and only if $n \notin A$. Then $n \notin A$ if and only if $g(n) \in A'$ if and only if $f(g(n)) \in B'$. As both g and f are computable, $\omega \setminus A$ is c.e. in B. □

1.2 The arithmetical hierarchy

Definition 1.3 (The relativized arithmetical hierarchy)**.** Let $X \subseteq \omega$ and $A \subseteq \omega$. Then for all $n \in \omega$

1. $\Delta^0_0(X) = \Sigma^0_0(X) = \Pi^0_0(X) = \{A \subseteq \omega : A \leq_T X\}$

2. A is $\Sigma^0_n(X)$ if there is an X-computable predicate R such that

$$x \in A \iff (\exists \vec{u}_1)(\forall \vec{u}_2)(\exists \vec{u}_3) \cdots (Q_n \vec{u}_n)\, R(x, \vec{u}_1, \ldots, \vec{u}_n),$$

where $Q_n = \exists$ if n is odd and $Q_n = \forall$ otherwise.

3. A is $\Pi^0_n(X)$ if there is an X-computable predicate R such that

$$x \in A \iff (\forall \vec{u}_1)(\exists \vec{u}_2)(\forall \vec{u}_3) \cdots (Q_n \vec{u}_n)\, R(x, \vec{u}_1, \ldots, \vec{u}_n),$$

where $Q_n = \exists$ if n is even and $Q_n = \forall$ otherwise.

4. A is $\Delta^0_n(X)$ if and only if $A \in \Sigma^0_n(X) \cap \Pi^0_n(X)$

We write Γ^0_n instead of $\Gamma^0_n(\emptyset)$ for $\Gamma \in \{\Sigma, \Pi, \Delta\}$.

The term arithmetical hierarchy comes from the fact that the Σ^0_n sets coincide with the sets definable in $(\mathcal{N}, 0, 1, +, \cdot)$ via $\exists_n$-formulas.

Proposition 1.5. *A set $A \subseteq \omega$ is Π^0_n if and only if $\omega \setminus A$ is Σ^0_n.*

We will now connect the arithmetic hierarchy with Turing reducibility and the Turing jump. Towards this note that we can iterate the Turing jump. Given $X \subseteq \omega$, let $X^{(0)} = X$ and for $n \in \omega$ let the **$(n+1)$-th Turing jump** of X be $X^{(n+1)} = (X^{(n)})'$.

Lemma 1.6. *For every $A, X \subseteq \omega$, $A \in \Sigma^0_1(X)$ iff A is c.e. in X if and only if $A \leq_m X'$.*

Proof. Proofs before, mutatis mutandis. □

Theorem 1.7 (Post's theorem)**.** *For every $A, X \subseteq \omega$ and $n > 0$, $A \in \Sigma^0_n(X)$ iff A is c.e. in $X^{(n-1)}$ iff $A \leq_m X^{(n)}$.*

Proof. The proof is by induction with the base case covered by Lemma 1.6. That A is c.e. in $X^{(n-1)}$ iff $A \leq_m X^{(n)}$ follows by replacing X in Lemma 1.6 with $X^{(n-1)}$.

Suppose by induction that the theorem holds for n, and suppose that A is $\Sigma^0_{n+1}(X)$ defined by the computable relation R. Then the set $P = \{(n, \vec{u}_1) : \forall \vec{u}_2 \ldots Q_n \vec{u}_n R(-)\}$ is $\Pi^0_n(X)$ and note that $A \in \Sigma^0_1(P)$. As $P \in \Pi^0_n(X)$, $\omega \setminus P$ is $\Sigma^0_n(X)$ and by the induction hypothesis, $\omega \setminus P \leq_m X^{(n)}$. In particular, $P \leq_T X^{(n)}$, and as A is c.e. in P, A is also c.e. in $X^{(n)}$.

On the other hand, suppose that A is c.e. in $X^{(n)}$. Then by the base case A is $\Sigma^0_1(X^{(n)})$, i.e. $x \in A \iff \exists u R(x, u)$ for R $X^{(n)}$-computable. Clearly $X^{(n)} \leq_m X^{(n)}$ and so by hypothesis $X^{(n)}$ is $\Sigma^0_n(X)$. Let Φ_e be such that $\Phi_e^X = R$ and let σ_k be the kth finite binary string in the lexicographic ordering. We claim that

$$x \in A \iff \exists u \exists k\ \Phi_e^{\sigma_k}(x, u) \downarrow = 1 \wedge (\forall m < |\sigma_k|)(\sigma_k(m) = 1 \leftrightarrow m \in X^{(n)}).$$

To see this recall the use principle. $\Phi_e^{X^{(n)}}(x, u) \downarrow = 1$ iff there is a $\sigma_k \prec X^{(n)}$ such that $\Phi_e^{\sigma_k}(x, u) \downarrow = 1$. $(\forall m < |\sigma_k|)\sigma_n(m) = 1 \leftrightarrow m \in X^{(n)}$ is a $\Sigma^0_{n+1}(X)$-definition, as $\Sigma^0_{n+1}(X) \supset \Sigma^0_n(X) \cup \Pi^0_n(X)$. Thus, the whole definition is $\Sigma^0_{n+1}(X)$. □

Corollary 1.8. *For $X, A \subseteq \omega$, A is $\Delta^0_n(X)$ if and only if $A \leq_T X^{(n-1)}$.*

Corollary 1.9. *For every $X \subseteq \omega$ and $n \in \omega$, there is $X^{(n)} \in \Sigma^0_n(X) \setminus \Pi^0_n(X)$.*

Definition 1.4. Say that X is **Γ^0_n-complete** for $\Gamma \in \{\Sigma, \Pi\}$ if every $Y \in \Gamma$, $Y \leq_m X$.

By Post's theorem we obtain that $\emptyset^{(n)}$ is Σ^0_n-complete.

1.3 Examples of complete sets

Recall that $(W_e)_{e\in\omega}$ is the canonical enumeration of c.e. sets, i.e., $W_e = \{n : \varphi_e(n)\downarrow\}$.

Proposition 1.10. *The set* $K_1 = \{e : W_e \neq \emptyset\}$ *is* Σ^0_1*-complete.*

Theorem 1.11. *The set* $\mathrm{Fin} = \{e : W_e$ *is finite*$\}$ *is* Σ^0_2*-complete and the set* $\mathrm{Tot} = \{e : W_e = \omega\}$ *is* Π^0_2*-complete.*

Proof. Suppose that $S \in \Sigma^0_2$, then $n \in S$ iff $\exists k \forall m R(n,k,m)$. Define $W_{f(n)}$ via a c.e. approximation as follows. For every s, let k_s be the least k such that $\forall m < s\ R(n,k,m)$ if such k exists and $k_s = s$ otherwise. Suppose we have defined $W_{f(n),s-1}$ and let j_{s-1} be the least element not in $W_{f(n),s-1}$. If $k_s > k_{s-1}$ enumerate j_{s-1} into $W_{f(n),s}$, otherwise $W_{f(n),s} = W_{f(n),s-1}$. We then let $W_{f(n)} = \lim_s W_{f(n),s-1}$. Note that every $W_{f(n),s}$ is computable, uniformly in s and that $W_{f(n),s} \supseteq W_{f(n),s-1}$ is increasing. Thus $m \in W_{f(n)}$ iff $\exists s m \in W_{f(n),s}$ and hence $W_{f(n)}$ is Σ^0_1, so c.e.

If $n \in S$, then $\lim_s k_s \downarrow$ and thus $W_{f(n)}$ is an initial segment of ω. On the other hand if $n \notin S$, then $\lim_s k_s \uparrow$ and furthermore for every m, there is s with $k_s = m$. Thus, $m \in W_{f(n)}$ and $W_{f(n)} = \omega$. □

1.4 Limit lemma

In the proof of Theorem 1.11 we have built a c.e. set as a limit of computable sets. Let us introduce the notion of limit computability more formally. A set A is **limit computable** in X if there is a sequence of finite sets $(F_i)_{i\in\omega}$, computable uniformly in X (i.e., there is an X-computable function f such that for all i,j $f(i,j) = \chi_{F_i}(j)$) such that $\chi_A = \lim_i \chi_{F_i}(j)$.

Theorem 1.12 (Limit lemma)**.** *A set* A *is* $\Delta^0_2(X)$ *(equivalently* $A \leq_T X'$*) if and only if there is a uniformly* X*-computable sequence of finite sets* $(F_i)_{i\in\omega}$ *such that for all* n*,* $A(n) = \lim_i F_i(n)$*.*

Proof. Suppose that $A \leq_T X'$, say $A = \Phi_e^{X'}$. Let X'_s be the stage s approximation of X', i.e., the finite set such that $n \in X'_s$ if and only if $\Phi_n^{X\cap\{0,\dots s\}}(n)\downarrow$. Clearly the sets X'_s are computably uniformly in X. Now let $A_s = \Phi_e^{X'_s}$, note that A_s is finite and computable in X, uniformly in s, as X'_s is a finite oracle (note that for this assumption to hold we not only need that X'_s is finite but also that $X'_s \subseteq \{0,\dots,s\}$). Now, if $\Phi_e^{X'} = i\downarrow$, then there is a shortest $\sigma \prec X'$ such that $\Phi_e^{\sigma}(i)\downarrow$. Furthermore, as X' is c.e., there is a stage s such that for all $t > s$, $X'_t \succeq \sigma$. Thus, for all such t, $A_t(n) = A(n)$.

On the other hand suppose that $A = \lim_s A_s$ with $(A_s)_{s\in\omega}$ uniformly computable in X. Then

$$\begin{aligned} n \in A &\iff \exists m(\forall k > m) n \in A_k \\ n \notin A &\iff \exists m(\forall k > m) n \notin A_k \end{aligned}$$

The first equation shows that A is $\Sigma^0_2(X)$ and the second equation shows that A is $\Pi^0_2(X)$. Thus, A is $\Delta^0_2(X)$. □

2 Cohen 1-generics

We will now start with our discussion of the computability theoretic version of Cohen forcing. The main subject of this section are Cohen 1-generics, to introduce them we do not need a forcing relation yet. The compatible forcing relation will be introduced in the next section.

Cohen invented the forcing method to prove the independence of the continuum hypothesis from ZFC. Since then many variations of forcing exist, and it has established itself as a central tool in mathematical logic. As the name suggests, a *Cohen generic real* $g \in 2^\omega$ is one that is typical and does not have any special properties. The term "special" here can be formalized by

stating that the set of elements having a property P is small in the sense of category, i.e., it is meager. We will discuss this point of view in more detail in a later section.

1-generics were isolated by Jockusch in the 80s even though they appear in many earlier computability theoretic constructions implicitly. We will work with 1-generic elements of 2^ω. However, our definitions and theorems can be easily adapted to ω^ω, and, in fact, any computably presented Polish space.

Definition 2.1. Let $R \subseteq 2^{<\omega}$ be closed upwards, i.e., $\sigma \in R$ implies $\tau \in R$ for all $\tau \succeq \sigma$. We say that $\gamma \in 2^{<\omega}$ **decides** R if

$$\gamma \in R \text{ or } (\forall \sigma \succeq \gamma)\sigma \notin R.$$

An element $g \in 2^\omega$ is **1-generic** if for every upwards-closed c.e. $R \subseteq 2^{<\omega}$, there is $\sigma \prec g$ such that σ decides R.

If R is not upwards-closed then we say that g decides R if it decides its upwards closure.

Definition 2.2. For g 1-generic we say that

1. $\sigma \prec g$ **forces** g **into** $R \subseteq 2^{<\omega}$ if $\sigma \in R$, in which case we also say that g **meets** R.
2. $\sigma \prec g$ **forces** g **out of** $R \subseteq 2^{<\omega}$ if $(\forall \tau \succeq \sigma)\tau \notin R$, in which case we also say that g **avoids** R.

Let us investigate the Turing degrees of 1-generic sets. 1-generics cannot be computable. To see this simply note that for computable $c \in 2^\omega$, the set $R_c = \{\sigma : \sigma \not\prec c\}$ is dense, thus if g is 1-generic, it must be forced into it and cannot be equal to c.

Proposition 2.1. *There is a 1-generic computable from* $\mathbf{0}'$.

Proof. We construct g in stages, i.e., $g \succ \ldots p_s \succ p_{s-1} \succ \ldots p_0$. Fix an enumeration of the c.e. sets $(W_e)_{e\in\omega}$. At stage s, having defined p_{s-1} ask whether there is $q \succeq p_{s-1}$ such that $q \in W_s$. If so, then let $p_s = q$, otherwise $p_s = p_{s-1}$.

Clearly, for every c.e. set W_e, p_e decides W_e. To see that g is indeed an infinite string note that the set $\{\sigma \in 2^{<\omega} : |\sigma| \geq n\}$ is c.e. and dense; thus, at infinitely many stages the length of p_s increases. To see that g is computable from $\mathbf{0}'$, note that at each stage we have to check whether

$$(\exists q \succ p_s)q \in W_e.$$

This is Σ^0_1, so $\mathbf{0}'$ decidable. □

Proposition 2.2. *Suppose that g is 1-generic and W is a c.e. set such that $W \leq_T g$. Then W is computable.*

Proof. Suppose that $W \leq_T g$ and $\Phi^g_e = W$. Consider the set

$$S = \{\sigma : \exists n(\Phi^\sigma_e(n) = 0 \wedge n \in W)\}.$$

This is a c.e. set and thus decided by an initial segment of g. As no initial segment of g can be in S, there is $\sigma \prec g$ such that for every $\tau \succ \sigma$, $\tau \notin S$. We now claim that

$$n \notin W \iff (\exists \tau \succ \sigma)\Phi^\tau_e(n) = 0$$

which would show that W is co-c.e. and thus computable. Certainly, as $g \in [\sigma]$ and $\Phi^g_e = W$, if $n \notin W$, then there is $\tau \succ \sigma$ such that $\Phi^\tau_e(n) = 0$. On the other hand, if $\Phi^\tau_e(n) = 0$ for some $\tau \succ \sigma$, then as $\tau \notin S$ $n \notin W$. Therefore, W is co-c.e., and thus computable. □

Definition 2.3. A degree $\mathbf{d}$ is **generalized low** if $\mathbf{d}' = \mathbf{d} \oplus \mathbf{0}'$ and it is **low** if $\mathbf{d}' = \mathbf{0}'$. Likewise, $x \in 2^\omega$ is (generalized) low if $deg(x)$ is (generalized) low.

Theorem 2.3. *Every 1-generic is generalized low.*

Proof. Consider the sets

$$J_e = \{\sigma : \Phi_e^\sigma(e) \downarrow\}.$$

The sets J_e are c.e. and thus decided by 1-generics, that is if g is 1-generic, then

$$\exists \sigma \prec g (\Phi_e^\sigma(e) \downarrow \vee (\forall \tau \succeq \sigma) \Phi_e^\tau(e) \uparrow) \tag{J_e}$$

Now, for a given string σ, $\emptyset'$ can check whether it satisfies (J_e) and which of the disjuncts it satisfies, i.e., whether it forces into J_e, or out. Thus using $g \oplus \emptyset'$ we can check whether $e \in g'$ by finding the least k such that $g \upharpoonright k$ witnesses (J_e). Thus $g \oplus \emptyset' \geq_T g'$. On the other hand, for any set x, $x' \geq_T \emptyset'$ (see Theorem 1.4), thus the equivalence follows. □

The statements (J_e) in the above proof are often referred to as **forcing the jump** requirements and a set x is said to force the jump if it satisfies all these requirements, i.e., decides all the sets J_e. Unsurprisingly, forcing the jump is a sufficient condition for 1-genericity.

Theorem 2.4. *A real $x \in 2^\omega$ is 1-generic if and only if x forces the jump.*

Proof. The left-to-right direction is trivially true as the sets J_e are c.e.. For the other direction define the computable function $f : \omega \to \omega$ for all n by

$$\Phi_{f(e)}^\sigma(n) = \begin{cases} 1 & \text{if } (\exists \tau \preceq \sigma)\tau \in W_{e,|\sigma|} \\ \uparrow & \text{otherwise} \end{cases}.$$

We then have that $\sigma \in J_{f(e)}$ iff $\sigma \in W_e$ and thus if $\tau \prec x$ decides $J_{f(e)}$ then τ decides W_e, showing that if x forces the jump then it is 1-generic. □

Theorem 2.5 (Friedberg jump inversion)**.** *For every $\mathbf{a} \geq \mathbf{0}'$, there is a 1-generic g with*

$$g' \equiv_T g \oplus \emptyset' \in \mathbf{a}.$$

Proof. We build g in stages interleaving forcing the jump requirements and coding $A \in a$. As usual $p_0 = \emptyset$. Suppose we have define p_{2s}, then let $p_{2s+1} = p_{2s} \frown A(s)$. Then, having defined p_{2s+1} look for the lexicographically least $\sigma \succeq p_{2s+1}$ forcing the jump J_s and let $p_{2s+2} = \sigma$. Let $g = \lim p_s$ and note that $n \in g'$ iff $\Phi_n^{p_{2n+2}}(n) \downarrow$. If $A \geq_T \emptyset'$, then A can check the latter condition and thus can compute g'. On the other hand, we can use $g \oplus \emptyset'$ to compute A inductively. To compute whether $n \in A$, knowing how to compute whether p_{2n-2}, use $\emptyset'$ to determine p_{2n-1} and note that $n \in A$ iff $g(|p_{2n-1}| + 1) = 1$. That $g \oplus \emptyset' \equiv_T g'$ follows from Theorem 2.3 as g forces the jump by construction. □

Corollary 2.6. *The jump $J : 2^\omega \to 2^\omega$ is onto $\{\mathbf{d} \geq \mathbf{0}'\}$.*

2.1 Relative 1-genericity and mutual genericity

Say that $g \in 2^\omega$ is **1-generic relative to** $x \in 2^\omega$, or x-1-generic, if for every upwards-closed x-c.e. $R \subseteq 2^{<\omega}$, g decides R. Furthermore, $x, y \in 2^\omega$ are **mutually 1-generic** if x is y-1-generic, and y is x-1-generic.

Note that if $x \equiv_T y$ and g is x-1-generic, then it is also y-1-generic by definition. Thus, the notion of a $\mathbf{d}$-1-generic for a Turing degree $\mathbf{d}$ is well-defined.

Proposition 2.7. *Suppose that g is x-1-generic. Then x is not c.e. in g, unless x is c.e.*

Proof. The proof is quite similar to the proof of Proposition 2.2. Suppose that x is c.e. in g, say $x = W_e^g$ and consider the set

$$S = \{\sigma : \exists n(n \in W_e^\sigma \wedge x(n) = 0\}.$$

S is c.e. in x and thus decided by g. In fact, g avoids S, as otherwise $n \in W_e$ and $x(n) = 0$, a contradiction. Thus, there is $\sigma \prec g$ forcing out of S. We now claim that

$$x(n) = 1 \iff (\exists \tau \succ \sigma) n \in W_e^\tau$$

which shows that x is c.e. If $x(n) = 1$, then as $W_e^g = x$, there is $\tau \succeq \sigma$ such that $n \in W_e^\tau$. On the other hand, i $n \in W_e^\tau$, then as $\tau \notin S$ by the genericity of g, $x(n) = 1$. □

In particular, if x is not computable, then $g \not\geq_T x$. This property of 1-generics is often referred to as **cone avoidance**, as for non-computable x and g x-1-generic, g avoids the upper cone of x, i.e., is not a member of the set $\{y \in 2^\omega : y \geq_T x\}$. A simple relativization of the fact that 1-generics cannot be computable, shows that x-1-generics also avoid the lower cone of x (i.e., they are not elements of the ideal $\{y \in 2^\omega : y \leq_T x\}$). However, by marginally adapting the proof of Proposition 2.7 we can show something much stronger.

Definition 2.4. Turing degrees $\mathbf{a}, \mathbf{b} \not\leq \mathbf{0}$ form a **minimal pair** if whenever $\mathbf{x} \leq \mathbf{a}$ and $\mathbf{x} \leq \mathbf{b}$, then $\mathbf{x} \leq \mathbf{0}$.

Theorem 2.8. *For every* $\mathbf{a} \not\leq \mathbf{0}$ *there is* $\mathbf{d}$ *such that* $\mathbf{a}$ *and* $\mathbf{d}$ *form a minimal pair.*

Exercise 2.9. Prove Theorem 2.8. *Hint: Modify the proof of Proposition 2.7 to obtain that if* $y \leq_T x$ *and* y *is c.e. in* g *which is* x*-1-generic, then* y *is c.e..*

Below we will need the computable join of strings in $2^{<\omega}$. We thus extend the definition two strings $\sigma, \tau \in 2^{<\omega}$ with $|\sigma| = |\tau|$ in the canonical way.

Theorem 2.10. *A real* $g \in 2^\omega$ *is 1-generic if and only if* $g = g_1 \oplus g_2$ *with* g_1 *and* g_2 *mutually 1-generic.*

Proof. Suppose that $g = g_1 \oplus g_2$ is 1-generic. We have to show for any upwards closed g_1-c.e. set $W = rg(\Phi^{g_1})$, that g_2 decides W. Let

$$R = \{\sigma \oplus \tau : \tau \in \Phi^\sigma \wedge |\sigma| = |\tau|\}.$$

If $\sigma \oplus \tau \prec g$ with $\sigma \oplus \tau \in R$, then $\tau \in \Phi^\sigma$ and thus g_2 is forced into the upwards closure of Φ^{g_1}. If no such $\sigma \oplus \tau$ exists, then by the 1-genericity of g, there are $\sigma' \oplus \tau' \prec g$ so that for all $\gamma \succ \sigma' \oplus \tau'$, $\gamma \notin R$ and hence for all $\delta \succ \tau'$, $\delta \notin \Phi^{g_1}$ and thus g_2 avoids Φ^{g_1}. It follows that g_2 is g_1-1-generic and similarly we may show that g_1 is g_2-1-generic.

For the other direction, let W be upwards closed c.e. We must show that g decides W, using the mutual genericity of g_1 and g_2. Define the g_1-c.e. set

$$R_2 = \{\sigma : g_1 \upharpoonright |\sigma| \oplus \sigma \in W\}.$$

Then g_2 decides R_2. If there is $\tau \prec g_2$ such that $\tau \in R_2$, then $g_1 \upharpoonright |\tau| \oplus \tau$ forces g into W. Otherwise, suppose there is $\tau \prec g_2$ such that no extension is in R_2. Define

$$R_1 = \{\sigma : (\exists \rho \succ \tau)\sigma \oplus \rho \in W\}.$$

Then g_1 decides R_1. g_1 cannot meet R_1, because if $\gamma \prec g_1$ with $\gamma \in R_1$, then $\gamma \oplus \rho \in W$ and ρ would extend τ and be in R_2, a contradiction. Thus, let $\gamma \prec g_1$ be such that no extension of it is R_1. Then for $\delta \succ \gamma$ and $\rho \succ \tau$, $\delta \oplus \rho \notin W$. So $\gamma \oplus \tau$ forces out of W. □

Exercise 2.11. Given $x \in 2^\omega$, let $x^{[n]} = \{i : \langle n, i\rangle \in x\}$. Show that if x is 1-generic, then for every $i \neq j$, $x^{[i]}$ and $x^{[j]}$ are mutually 1-generic.

Corollary 2.12. *There is a minimal pair of degrees computable from* $\mathbf{0}'$.

Theorem 2.10 is a special version of the product forcing lemma in the general theory of forcing that says that if $\mathbb{P}$ and $\mathbb{Q}$ are two forcing partial orders and $G \subseteq \mathbb{P} \times \mathbb{Q}$ is a $\mathbb{P} \times \mathbb{Q}$ filter then G is M-generic if and only if G_1 is M-generic and G_2 is $M[G_1]$-generic.

2.2 Interlude: Category and genericity

Recall the usual product topology on Cantor space 2^ω having a basis of clopen sets

$$[\![\sigma]\!] = \{x \in 2^\omega : x \succ \sigma\}.$$

This topology is compatible with the canonical uniform probability measure defined by $\mu([\![\sigma]\!]) = 2^{-|\sigma|}$. Cantor space is the canonical example of a compact 0-dimensional Polish (completely metrizable, separable) space.

There is a natural correspondence between open sets in Cantor space and the upwards closed subsets $R \subseteq 2^{<\omega}$ which we used in our definition of 1-genericity:

1. For $U = \bigcup [\![\sigma_i]\!]$ let $R_U = \{\tau : \exists i\ \tau \succ \sigma_i\}$, then R_U is an upwards closed subset of $2^{<\omega}$.
2. For $R \subseteq 2^{<\omega}$ let $[\![R]\!] = \bigcup_{\sigma \in R} [\![\sigma]\!]$, then $[\![R]\!]$ is open.

Having established the correspondence between upwards-closed sets of $2^{<\omega}$ and open sets of 2^ω, we also get the dual: The downwards closed subsets of $2^{<\omega}$ correspond with closed sets in 2^ω. Recall that downwards closed subsets of $2^{<\omega}$ are binary branching trees. This insight is probably even more important than its dual.

Proposition 2.13. *A set $C \subseteq 2^\omega$ is closed if and only if there is a tree $T \subseteq 2^{<\omega}$ such that $C = [T] = \{x : (\forall \sigma \prec x)\sigma \in T\}$.*

Definition 2.5. A set $U \subseteq 2^\omega$ is **effectively open** if there is an upwards closed c.e. $R \subseteq 2^{<\omega}$ such that $U = [\![R]\!]$. Dually, it is **effectively closed** if there is a co-c.e. tree $T \subseteq 2^{<\omega}$ such that $U = [T]$.

Exercise 2.14. Show that for any co-c.e. tree $T \subseteq 2^{<\omega}$, there is a computable tree $\hat{T}$ with $[T] = [\hat{T}]$.

Exercise 2.15. Show that there are computable enumerations of the effectively open sets and effectively closed sets.

We can now extend our definition of meeting and avoiding sets in the canonical way. For $U \subseteq 2^\omega$ open, $U = [\![R]\!]$, $x \in 2^\omega$ **meets** U if there is $\sigma \prec x$ so that $\sigma \in R$. It **avoids** U if $\exists \sigma (\forall \tau \succ \sigma)\tau \notin R$.

Proposition 2.16. *For $g \in 2^\omega$, g is 1-generic if and only if g meets or avoids every effectively open set.*

We can extend the definition of meeting/avoiding to arbitrary sets $S \subseteq 2^\omega$. It turns out that the straightforward extension of meeting from open to arbitrary sets does not really correspond to the concept of genericity. After all, there is no set $R \subseteq 2^{<\omega}$ corresponding with S if S is not open. Thus, we say that

Definition 2.6. Let $\sigma \in 2^{<\omega}$ and $S \subseteq 2^\omega$. Then

- σ **forces membership** in S if $[\![\sigma]\!] \subseteq S$;

- σ **forces nonmembership** in S if $[\![\sigma]\!] \cap S = \emptyset$.

If $\sigma \prec x \in 2^\omega$ forces membership, then we say that x is **forced into** S by σ and if it forces nonmembership we say that x is **forced out of** S by σ.

Recall that for $S \subseteq 2^\omega$, the **interior of** S, $int(S)$, is the maximal open $U \subseteq 2^\omega$ such that $U \subseteq S$. Dually, the **closure of** S, $cl(S)$, is the minimal closed C such that $C \supseteq S$. Finally, $S \subseteq 2^\omega$ is **dense** if $cl(S) = 2^\omega$ and S is **nowhere dense** if $int(cl(S)) = \emptyset$. Clearly, S is nowhere dense if and only if $2^\omega \setminus S$ is dense.

To continue, a set $S \subseteq 2^\omega$ is **meager** if $S = \bigcup N_i$ with N_i nowhere dense, it is **comeager** if $2^\omega \setminus S$ is meager, and S is **nonmeager** if S is not meager.

Recall the famous Baire category theorem. Here is one of its many equivalent formulations.

Theorem 2.17 (Baire category theorem)**.** *Let $\mathcal{X}$ be a complete metric space and $D_i \subseteq \mathcal{X}$ dense for all D_i. Then $\bigcap D_i$ is dense.*

Note that the Baire category theorem immediately implies the existence of 1-generics without having to resort to constructing one explicitly.

Exercise 2.18. Let $(T_i)_{i\in\omega}$ be an effective enumeration of the computable trees. Say that $S \subseteq 2^\omega$ is Σ^0_2 if it is of the form $\bigcup_i [T_{f(i)}]$ for f a computable function. Show that if g is 1-generic then it is forced out of every meager Σ^0_2 set.

Exercise 2.19. Let $(\sigma_i)_{i\in\omega}$ be an effective enumeration of all binary strings. Say that a set $S \subseteq 2^\omega$ is Π^0_2 if it is of the form $\bigcap_i [\![\sigma_{f(i)}]\!]$ for f a computable function. Show that there is a 1-generic g and a meager Π^0_2 set P so that g is not forced out of g. *Hint: Use that there is a Δ^0_2 1-generic and use it to obtain a Π^0_2 set whose only member is g.*

2.3 Perfect antichains

Definition 2.7. A tree $T \subseteq 2^{<\omega}$ is **perfect** if it is non-empty and for every $\sigma \in T$, there are incompatible $\tau, \rho \in T$ with $\sigma \prec \tau, \rho$.

We extend this definition to $C \subseteq 2^\omega$ by saying that C is **perfect** if there is a perfect tree T with $[T] = C$. In other words, a perfect set C is a closed non-empty set without isolated points, i.e., $|C \cap [\![\sigma]\!]| \neq 1$ for every $\sigma \in 2^{<\omega}$. It is not hard to see that C is perfect if and only if it is homeomorphic to 2^ω. In particular, perfect sets have cardinality $2^{\aleph_0}$.

Lemma 2.20. *Suppose that $(D_i)_{i\in\omega}$ is a sequence of dense subsets of $2^{<\omega}$. Then there exists a perfect tree T such that for any $x \neq y \in [T]$ and any $i \in \omega$, there are $\sigma \prec x$ and $\tau \prec y$ so that $\sigma \oplus \tau \in D_i$.*

Proof. First note that for any sequence of strings $\sigma_0, \ldots, \sigma_i$ we can find extensions $\tau_0, \ldots, \tau_i$, $\tau_j \succ \sigma_j$ so that for given D_n, $\tau_j \oplus \tau_k$ meets D_n for $j, k \leq i$. To see this simply start with the given sequence $\sigma^0_0 = \sigma_0, \ldots, \sigma^0_i = \sigma_i$ and extend the strings in stages so that at stage $s = \langle j, k\rangle$ you ensure that $\sigma^{\langle j,k\rangle}_l \succeq \sigma^{\langle j,k\rangle - 1}_l$ for all l and if $j \neq k$, $\sigma^{\langle j,k\rangle}_j \oplus \sigma^{\langle j,k\rangle}_k \in D_n$. Supposing without loss of generality that $\langle i, i\rangle > \langle j, k\rangle$ for every j, k with $j < i$ or $k < i$, we let $\tau_j = \sigma^{\langle i,i\rangle}_j$ for $j < i$ giving the desired sequence.

We will define T as the limit of the downwards-closures of the finite sets S_s. We have $S_0 = \emptyset$ and having defined S_s, let $S' = \{\sigma^\frown 0 : \sigma \in S_s\} \cup \{\sigma^\frown 1 : \sigma \in S_s\}$ and S_{s+1} be the set obtained from S' using the procedure in the above paragraph with the set $\bigcap_{i\leq s} D_i$ and the elements of S' listed in lexicographic order. This finishes the construction.

Clearly T is perfect. Say that $x, y \in [T]$ and $i \in \omega$. Then by construction there is $\sigma \prec x$ and $\tau \prec y$ and $s > i$ so that $\sigma, \tau \in S_s$. Then $\sigma \oplus \tau \in D_i$ by construction. □

Definition 2.8. A set of Turing degrees $A \subseteq \mathcal{D}_T$ is an **antichain** if for every $\mathbf{a}_0, \mathbf{a}_1 \in A$, $\mathbf{a}_0 | \mathbf{a}_1$.

Theorem 2.21. *There exists a perfect sets of pairwise mutual 1-generics and thus a perfect antichain.*

Proof. Use Lemma 2.20 with the collection D_i of dense sets being the c.e. and co-c.e. upwards-closed dense sets. Then for elements $x, y \in [T]$ $x \oplus y$ is 1-generic. So, by Theorem 2.10 they are mutually 1-generic, hence $x \not\geq_T y$ and $y \not\geq_T x$. This gives the required perfect antichain. □